Fixed-Size Traffic: Iscandar vs Buddy

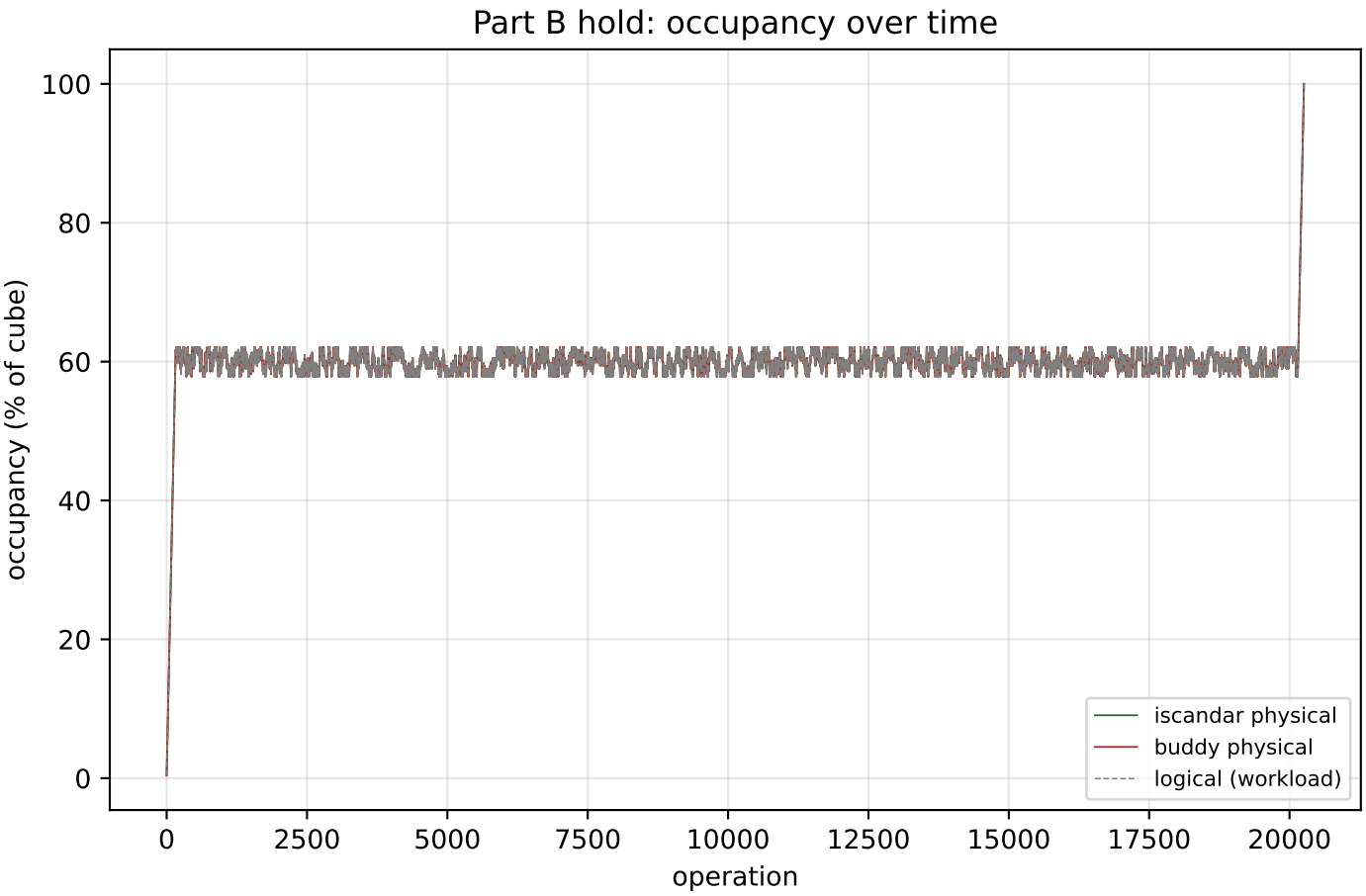
```
Load: Fixed-Size Traffic (author's shelf,
  2026-10-26): every request exactly 128 sectors,
  like a protocol stack's message blocks. Zero
  size variance is the buddy system's best
  possible world: the fairness control.

Engines: the full Iscandar engine (all modes,
  exact fit) vs the classic binary buddy.
Arena: 32,768 sectors (128 KiB at 4 B/sector).
The same seeded stream drives both engines.

Part A - the sprint (fill to 85% logical, churn,
push; first failed allocation loses):
  buddy    stalled op 1384 (churn) at 100.00% physical
  iscandar stalled op 1384 (churn) at 100.00% physical
  margin 0 ops to iscandar.

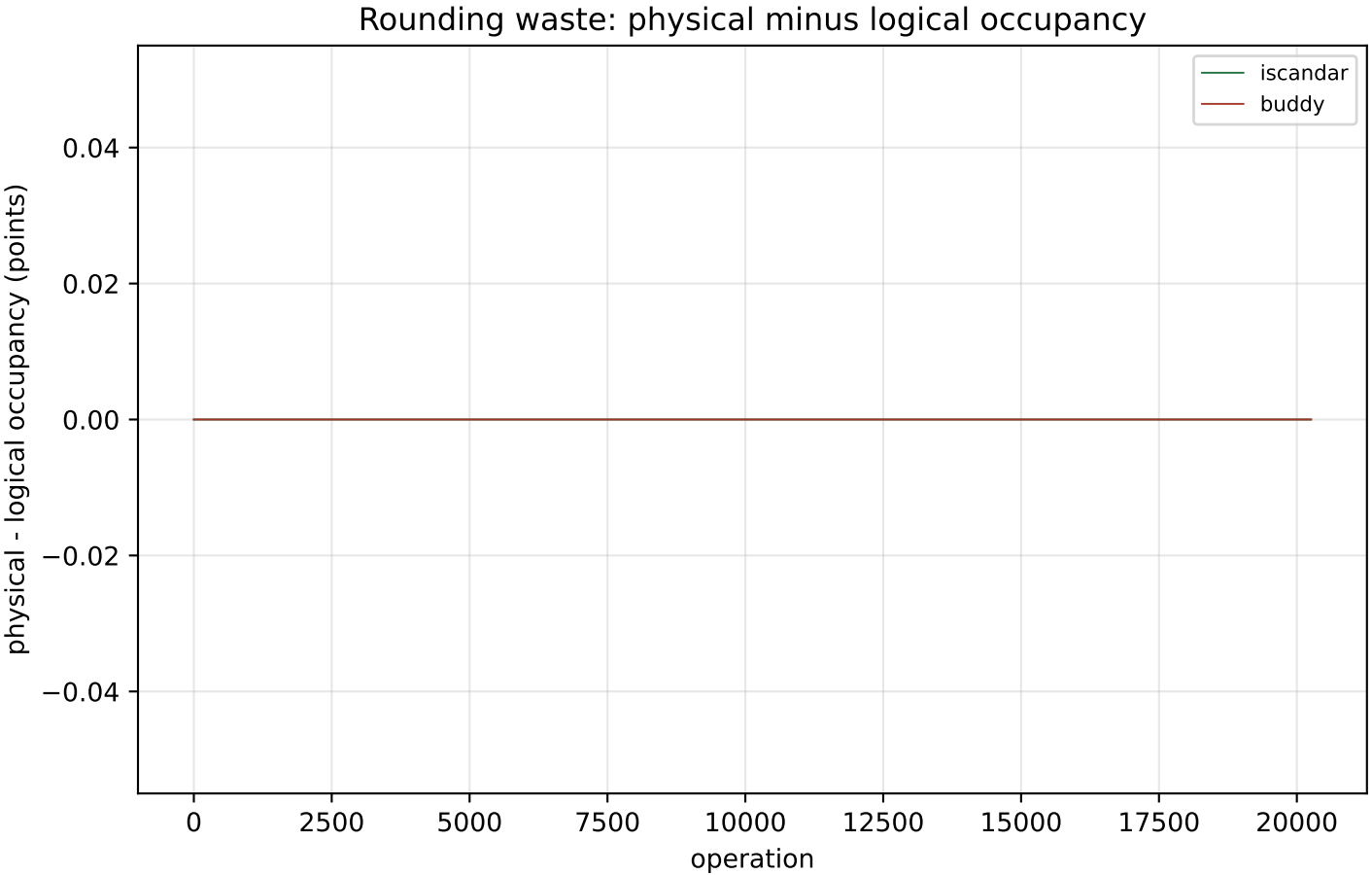
Part B - the steady-state hold (fill to 60%
logical, 20000 ops in a +/-2-point band, push):
  refusals during the hold: iscandar 0, buddy 0
  push stall: buddy at 100.00% physical, largest
           free block 0 sectors
  push stall: iscandar at 100.00% physical, largest
           free run 0 sectors

GammaCheck green in both parts (block checks
every op, full-ladder checks at boundaries).
```


Part B hold: occupancy over time
occupancy (% of cube)
operation
0
20
40
60
80
100
0
2500
5000
7500
10000
12500
15000
17500
20000
iscandar physical
buddy physical
logical (workload)

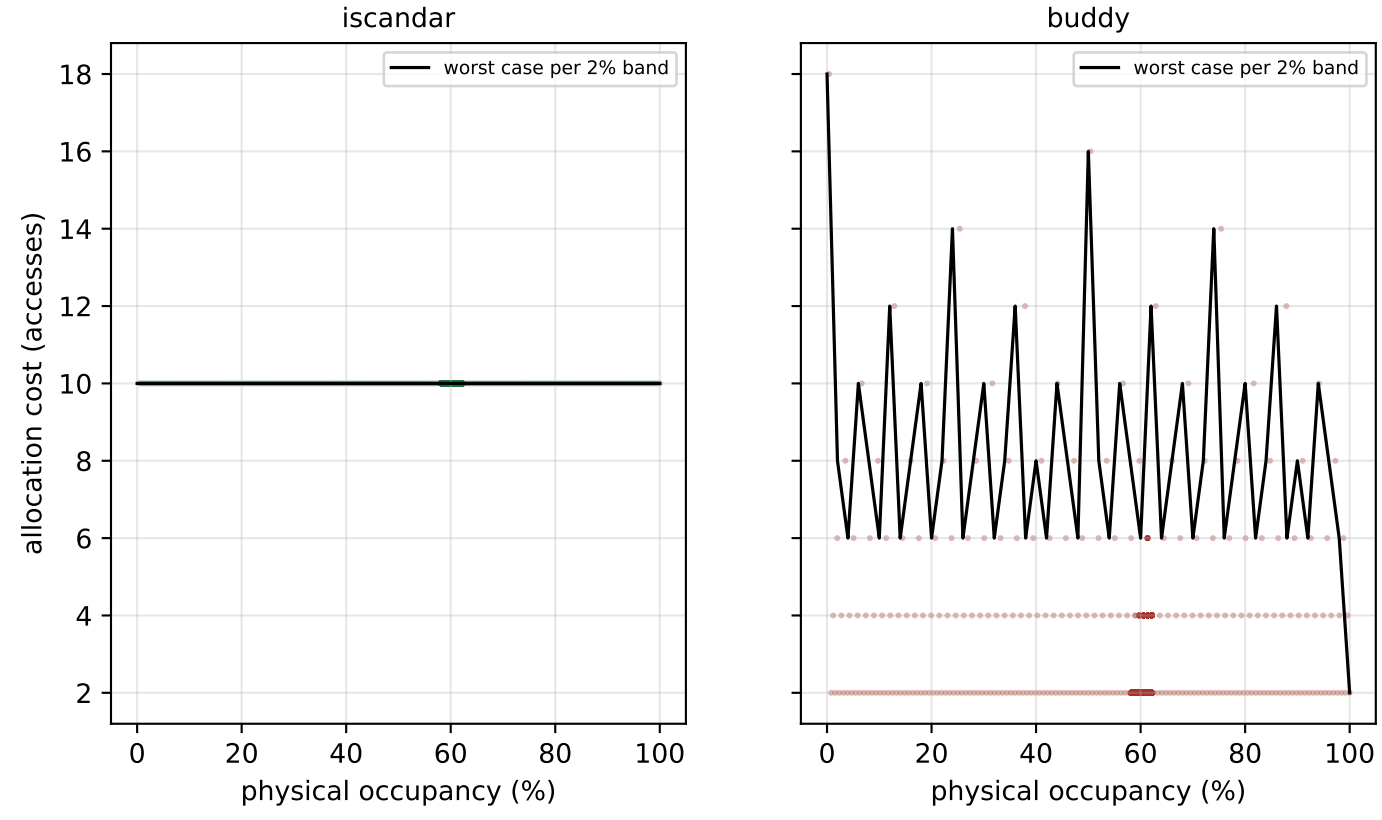
Commentary - occupancy over time

The gray dashed line is the workload itself, held at 60% logical. Iscandar's green line lies on top of it (mean 60.0% physical), because exact fit makes physical occupancy equal to logical occupancy. Buddy's red line rides above it: mean 60.0% physical, ranging 57.8 to 62.1%. The vertical distance between the red line and the workload is space Buddy has consumed without holding data.


Rounding waste: physical minus logical occupancy
iscandar
buddy
physical - logical occupancy (points)
0.04
0.02
0.00
−0.02
−0.04
0
2500
5000
7500
10000
12500
15000
17500
20000
operation

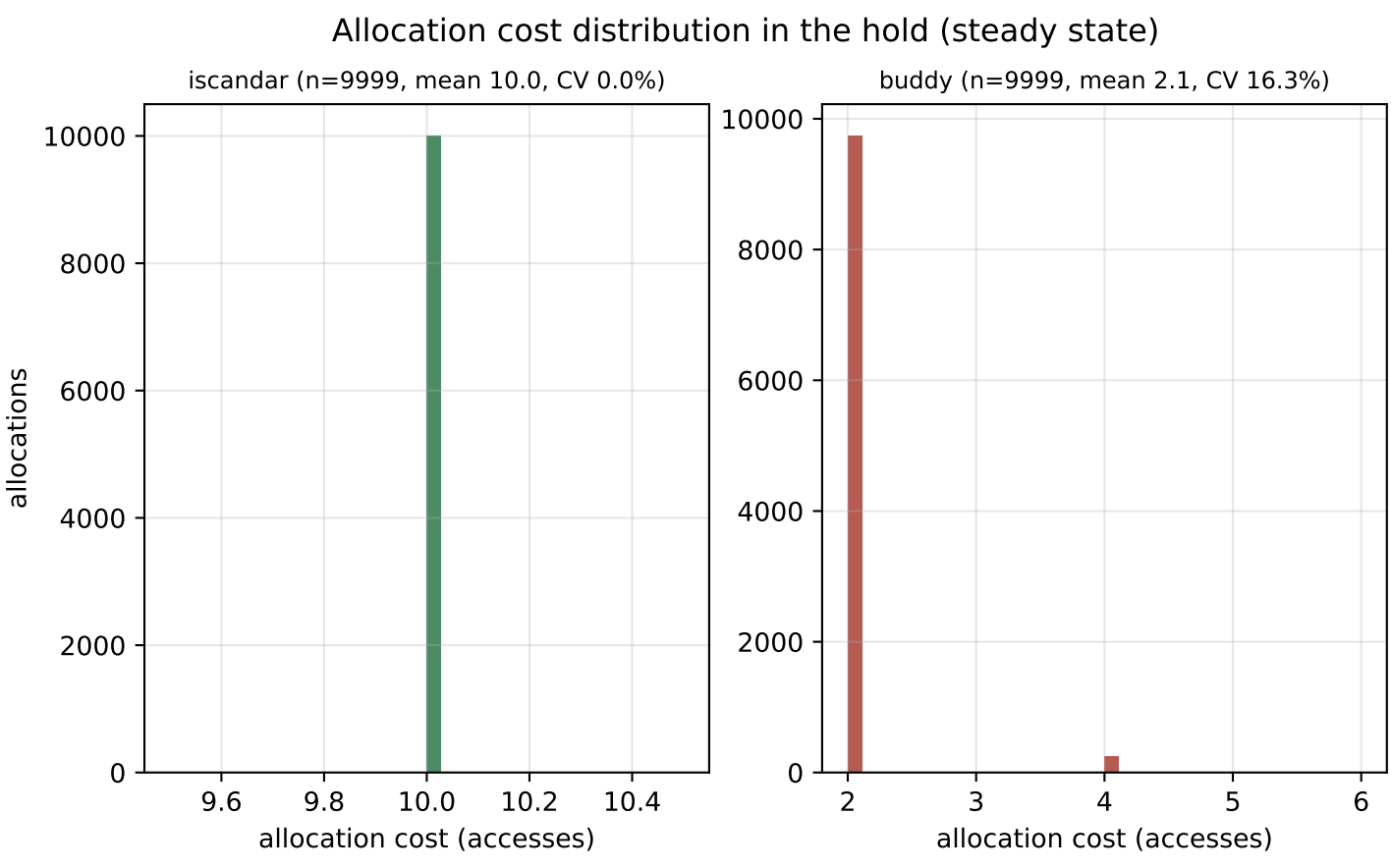
Commentary - rounding waste

Both lines lie on zero for the entire run: under this load there is no rounding waste on either side. Buddy's block size equals the request size, so its lattice wastes nothing, and exact fit wastes nothing by construction. That is the fairness control behaving as designed - this profile is the one case where the two engines' occupancy stories are identical.


Allocation cost against occupancy (hold + push; units differ per engine, see closing page)
iscandar
buddy
worst case per 2% band
worst case per 2% band
allocation cost (accesses)
physical occupancy (%)
physical occupancy (%)
2
4
6
8
10
12
14
16
18
0
20
40
60
80
100

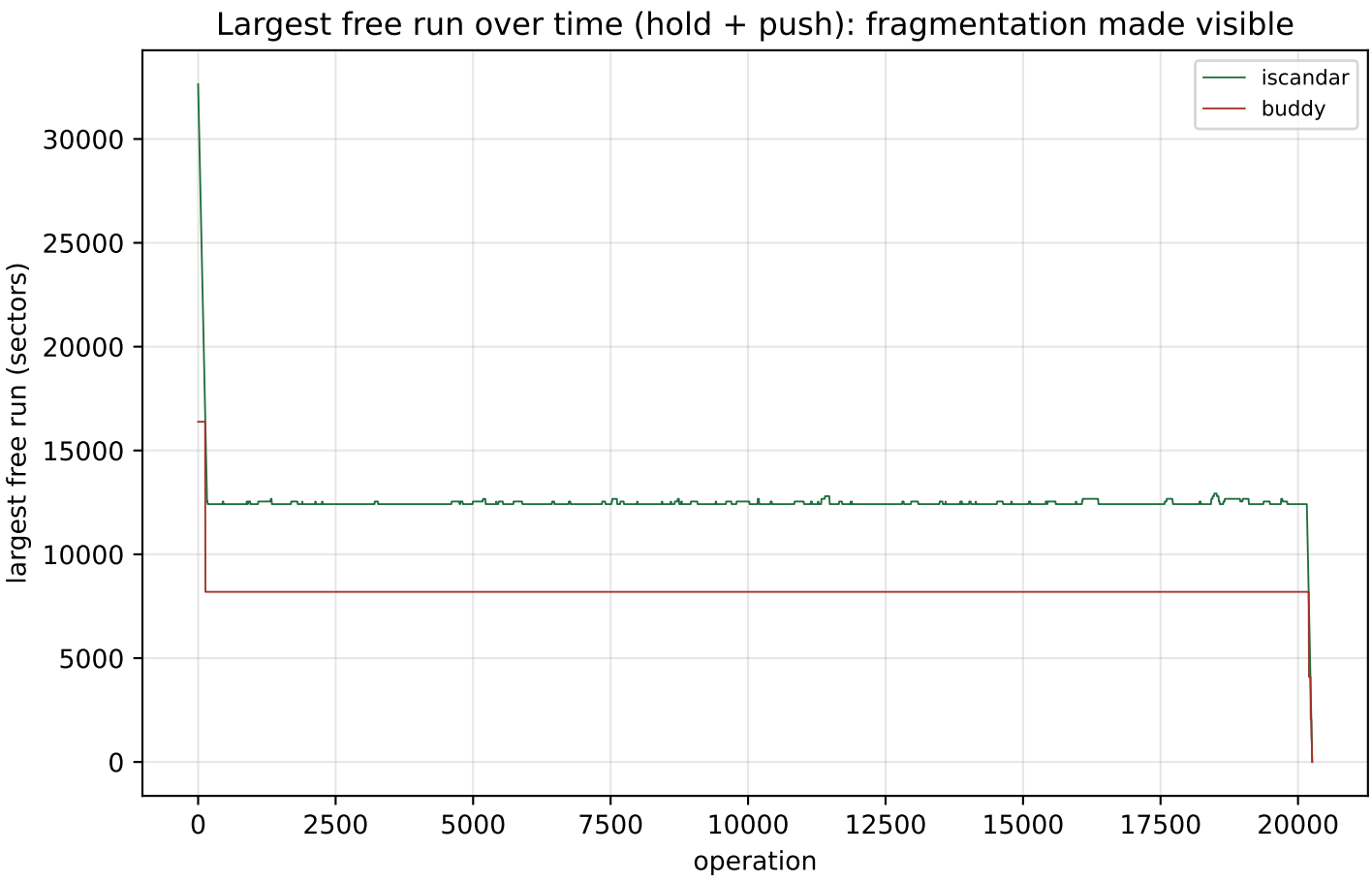
Commentary - cost against occupancy

Two panels with a black worst-case envelope. Iscandar's envelope stays between 10 and 10 touches across every occupancy band it visits; the price does not blow up as the cube fills. Buddy's envelope stays between 2 and 18 in its own tree-visit units (the levels are not comparable across panels; the shapes are). The readable point is the absence of a cliff: neither engine's worst case runs away with occupancy under this load.


Allocation cost distribution in the hold (steady state)
iscandar (n=9999, mean 10.0, CV 0.0%)
buddy (n=9999, mean 2.1, CV 16.3%)
allocations
0
2000
4000
6000
8000
10000
9.6
9.8
10.0
10.2
10.4
2
3
4
5
6
allocation cost (accesses)
allocation cost (accesses)

Commentary - cost distributions

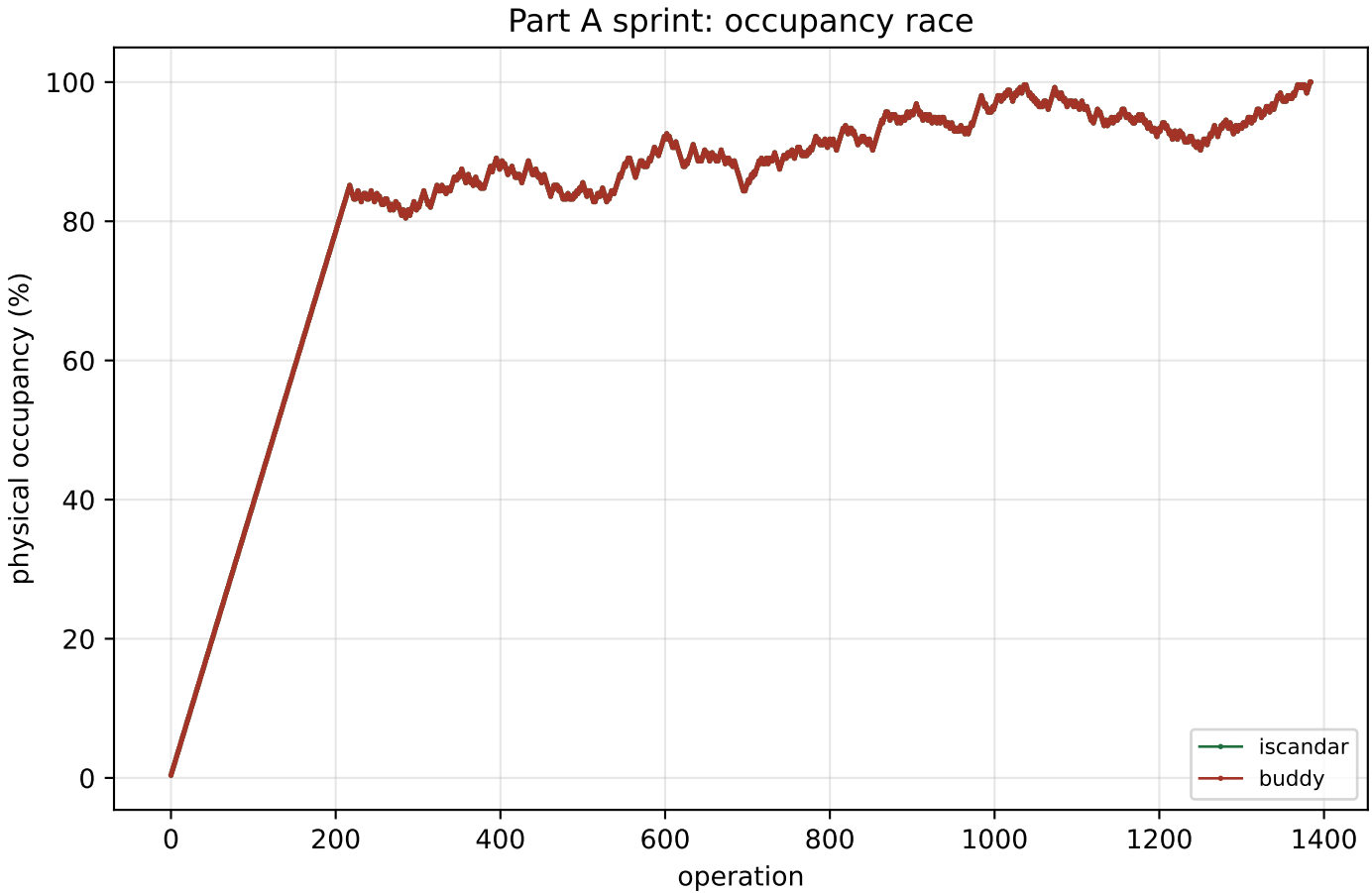
In the hold, Iscandar's allocation cost has median 10, with the middle half between 10 and 10 and the 90th percentile at 10. Buddy's has median 2, middle half 2 to 2, 90th percentile 2. The stability sigma on the closing page states the same shapes as a single number per series.


Largest free run over time (hold + push): fragmentation made visible
iscandar
buddy
largest free run (sectors)
30000
25000
20000
15000
10000
5000
0
0
2500
5000
7500
10000
12500
15000
17500
20000
operation

Commentary - largest free run

The fragmentation picture. Through the hold, Iscandar's largest contiguous free run has median 12416 sectors; Buddy's largest free block has median 8192. At the push stalls, Buddy dies with a largest free block of 0 sectors at 100.00% physical, Iscandar with a largest free run of 0 at 100.00% physical.

Graph 5 is the one that makes the RTOS applications engineer stop and look. The occupancy graphs show what Buddy wastes; the largest-free-run graph shows what that waste costs - the size of the biggest promise each allocator can still keep.


Part A sprint: occupancy race
physical occupancy (%)
operation
0
20
40
60
80
100
0
200
400
600
800
1000
1200
1400
iscandar
buddy

Commentary - the sprint

buddy stalls first, at op 1384 (100.00% physical); iscandar continues alone and stalls at op 1384 (100.00% physical). The margin is 0 ops to iscandar.

Numbers and caveats

```
Numbers (population statistics over successful ops)

sprint iscandar alloc: n=820 mean 10.00 var 0.00 CV 0.0% stability inf sigma, range 10..10

sprint buddy alloc: n=820 mean 2.67 var 3.31 CV 68.2% stability 1.5 sigma, range 2..18

hold   iscandar alloc: n=9999 mean 10.00 var 0.00 CV 0.0% stability inf sigma, range 10..10
hold   iscandar free:  n=10001 mean 6.00 var 0.00 CV 0.0% stability inf sigma, range 6..6

hold   buddy alloc: n=9999 mean 2.05 var 0.11 CV 16.3% stability 6.1 sigma, range 2..6
hold   buddy free:  n=10001 mean 2.03 var 0.03 CV 8.2% stability 12.2 sigma, range 2..4

Counting and caveats

- Iscandar accesses are sector-word touches on
  the engine path: probe words in the search plus
  the commit's word touches, per the unified
  model in gaussian_race.py's header. Gate
  (counter) reads are a separate series and are
  not blended into accesses.
- Buddy accesses are its tree-work analog units
  (buddy.py), NOT the same physical units. Read
  across engines by shape, CV, and outcome;
  read levels and sigmas only within one engine
  against its own bound (the two-series
  discipline adopted 2026-10-26).
- Stability sigma is mean / SD (the reciprocal
  of CV): how many of its own standard
  deviations the mean cost sits above zero. It
  is a stability ratio, not a defect-rate
  sigma; a defect sigma needs a declared limit.
- Placement completeness (post-amendment): an
  Iscandar failure means no contiguous free run
  of the request existed, or the request
  exceeded all free space. It never fails facing
  an exact-fit hole.
- One seeded stream drove both engines in each
  part (seed 1). GammaCheck green throughout.
```